Exercise 5

Donnerstag, 27. November 2025 08:04

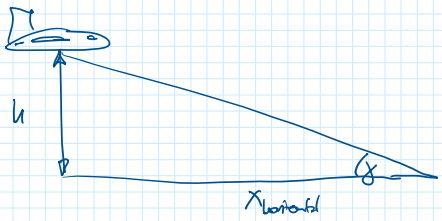
Exercise 5: Turning Flight

A turbojet aircraft (symmetric quadratic drag polar) begins to glide at a height h_0=1000 m above the ground.

Zero-lift drag coefficient	C_{D0}	0.031
K-factor	k	0.026
Reference area	S_{ref}	18 m²
Aircraft mass	m	2000 kg
Maximum engine performance parameter	L_P	6000

Height	h	1000 m
Density	ρ	$1.1117 \frac{kg}{m^3}$

a) What is the maximum horizontal distance the aircraft can cover?



(1) $\frac{\Delta h}{\Delta x} = \tan(\gamma)$

(2) $\varepsilon = \frac{c_D}{c_L} = -\tan(\gamma)$

Max horizontal distance @ ε^* → Symmetric quadratic drag polar

→ $\varepsilon^* = 2\sqrt{k \cdot c_{D0}}$

(1) & (2): $\Delta x_{max} = \frac{\Delta h}{-\varepsilon^*} = \frac{-1000m}{-2\sqrt{0{,}026 \cdot 0{,}031}} = 17666m$

A320: $E = 20 \rightarrow \varepsilon^* = \frac{1}{E} = 0.05$

@ $h = 12000m$

$\Delta x = \frac{-12000m}{-0.05} = 240\,km$;

Derivation of *-quantities:

PM-Equations: $\dot{V} = \frac{T-D}{m} - g \cdot \sin\gamma \overset{!}{=} 0$ (0, glide!)

$\dot{\chi} = \frac{L \cdot \sin\mu}{mV\cos\gamma} \overset{!}{=} 0$

$\dot{\gamma} = \frac{L \cdot \cos\mu}{mV} - \frac{g}{V}\cos\gamma \overset{!}{=} 0$

$\Rightarrow D = -mg\sin\gamma$, $\mu = 0$, $L = mg\cos\gamma$

$\frac{-D}{L} = \frac{mg\sin\gamma}{mg\cos\gamma} = \tan\gamma = -\frac{\frac{1}{2}\rho V^2 S_{ref} \cdot c_D}{\frac{1}{2}\rho V^2 S_{ref} \cdot c_L} = -\frac{c_D}{c_L}$

$\Rightarrow \varepsilon = -\tan\gamma = \frac{c_D}{c_L}$; Fly at *-values: c_D^*, c_L^*, V^* →

$\varepsilon = \frac{c_D}{c_L} = \frac{c_{D0} + k \cdot c_L^2}{c_L} = \frac{c_{D0}}{c_L} + k \cdot c_L$

$\frac{\partial \varepsilon}{\partial c_L} \overset{!}{=} 0 : -\frac{c_{D0}}{c_L^2} + k = 0 \rightarrow c_L^* = \sqrt{\frac{c_{D0}}{k}}$

$c_D^* = c_{D0} + k \cdot c_L^2 = c_{D0} + \frac{c_{D0}}{k} \cdot k = 2c_{D0}$

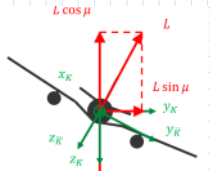
$V^* = \sqrt{\frac{2mg}{\rho S_{ref} c_L^*}} = \sqrt{\frac{2mg}{\rho S_{ref}\sqrt{\frac{c_{D0}}{k}}}}$

$\varepsilon^* = 2\sqrt{k \cdot c_{D0}}$

b) What are the control inputs to the point mass model and how can the pilot influence them?

$\dot{V} = \frac{T-D}{m} - g\sin(\gamma_K)$

$\dot{\chi} = \frac{L\sin\mu_K}{mV\cos(\gamma_K)}$



States: V → T

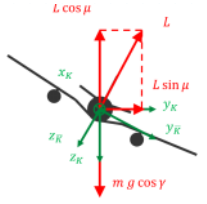
γ → L

χ → μ

$$\dot{V} = \frac{T-D}{m} - g\sin(\gamma_K)$$

$$\dot{\chi} = \frac{L\sin\mu_K}{m\,V\cos(\gamma_K)}$$

$$\dot{\gamma} = \frac{L\cos\mu_K}{m\,V} - \frac{g}{V}\cos(\gamma_K)$$



States: $V \rightarrow I$

$\gamma \rightarrow L$

$\chi \rightarrow \mu$

- Direction is governed by the direction of $L \rightarrow \mu$
- Magnitude of L/D : c_L
- Magnitude of Thrust : δ_T

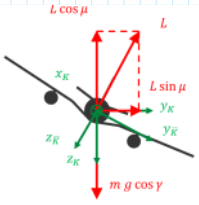
Control Input	Influence	Pilot
δ_T	Thrust	Thrust Lever
c_L	Magnitude Lift	Surface via Stick
μ_K	Direction of Lift	Surfaces via Stick

c) Derive the required velocity for achieving the maximum climb angle. Which climb angle and which velocity is reached?

$$\dot{V} = \frac{T-D}{m} - g\sin(\gamma_K)$$

$$\dot{\chi} = \frac{L\sin\mu_K}{m\,V\cos(\gamma_K)}$$

$$\dot{\gamma} = \frac{L\cos\mu_K}{m\,V} - \frac{g}{V}\cos(\gamma_K)$$



Boundary Conditions:

$\dot{\chi} = 0$

$\dot{\gamma} = 0,\ \gamma = \text{const.}$

$\dot{V} = 0$

Plug in Boundary Condition in PM Equations:

$$\left.\begin{aligned} T-D &= mg\cdot\sin\gamma \\ \mu &= 0 \\ L &= mg\cdot\cos\gamma \end{aligned}\right\} \quad \tan\gamma = \frac{T-D}{L}, \quad \sin\gamma = \frac{T-D}{mg}$$

$$\sin\gamma = \frac{T-D}{mg} \rightarrow \frac{T}{mg} = \sin\gamma + \frac{D}{mg} \Rightarrow \frac{T}{mg} = \sin\gamma + \frac{\varepsilon\cdot mg\cos\gamma}{mg} \rightarrow \boxed{\frac{T}{mg} = \sin\gamma + \varepsilon\cdot\cos\gamma}$$

$$\frac{D}{L} = \frac{c_D}{c_L} = \varepsilon \rightarrow D = \varepsilon\cdot L = \varepsilon\cdot mg\cos\gamma$$

Quadratic Symmetric Drag Polar: $c_D = c_{D_0} + k\cdot c_L^2$

$$\rightarrow \sin\gamma = \frac{T}{mg} - \frac{c_D}{c_L}\cos\gamma = \frac{T}{mg} - \left(\frac{c_{D_0}}{c_L} + k\cdot c_L\right)\cos\gamma$$

$$\sin\gamma = \frac{T}{mg} - \frac{c_{D_0}}{c_L^*}\left(\frac{V}{V^*}\right)^2 - k\cdot c_L^*\left(\frac{V^*}{V}\right)^2\cos^2\gamma$$

$$\sin\gamma = \frac{T}{mg} - \frac{\varepsilon^*}{2}\left(\frac{V}{V^*}\right)^2 - \frac{\varepsilon^*}{2}\left(\frac{V^*}{V}\right)\cos\gamma$$

$c_L^* = \sqrt{\frac{c_{D_0}}{k}}$, $k\cdot c_L^* = \frac{\varepsilon^*}{2}$, $\varepsilon^* = \frac{2c_{D_0}}{c_L^*} \rightarrow \frac{c_{D_0}}{c_L^*} = \frac{\varepsilon^*}{2}$

Recap a): $V^{*2} = \frac{2mg}{\rho S c_L^*} \rightarrow \left(\frac{V}{V^*}\right)^2 = \frac{\frac{2mg}{\rho S c_L}\cdot\cos\gamma}{\frac{2mg}{\rho S c_L^*}}$ (from PM Eq.)

$$\rightarrow c_L = \left(\frac{V^*}{V}\right)^2 c_L^*\cdot\cos\gamma$$

$$\boxed{\frac{T}{} = \frac{\varepsilon^*}{}\left[\left(\frac{V}{}\right)^2 + \left(\frac{V^*}{}\right)^2\cdot\cos^2\gamma\right]}$$

$$\frac{T}{mg} = \sin\gamma + \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2 \cos^2\gamma\right]$$

Back to the question: max climb angle γ_{max} & corresponding velocity

$\cos^2(\gamma) \approx 1$

$$\gamma(V) = \arcsin\left(\frac{T}{mg} - \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\right]\right)$$

$\max(\arg) \overset{!}{=} \max(\arcsin(\arg))$

$\arg = f(V)$

$$\frac{\partial f(V)}{\partial V} \overset{!}{=} 0 \qquad f(V) = \frac{T}{mg} - \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\right]$$

does not depend on V!

$$\gamma(V) = \arcsin\left(\frac{T-D}{mg}\right);$$

$$\sin\gamma = \frac{T-D}{mg} = \frac{T}{mg} - \frac{D}{mg} \quad \left.\right\} \rightarrow L = V\cdot\left(\frac{T}{mg} - \frac{D}{mg}\right)$$

$L = V\sin\gamma$

$$\frac{\partial f}{\partial V} = 0 = -\frac{2\varepsilon^*}{2}\frac{V}{V^{*2}} + \frac{2\varepsilon^*}{2}\frac{V^{*2}}{V^3}$$

$$\Rightarrow \frac{V}{V^{*2}} = \frac{V^{*2}}{V^3} \rightarrow V = V^*;$$

Compute V^*:

$$V^* = \sqrt{\frac{2mg}{\rho S_{ref} c^*}} = \sqrt{\frac{2mg}{\rho S_{ref}\sqrt{\frac{c_{D_0}}{k}}}} = 42.39\ m/s;$$

$$T = \delta_T \cdot L_{plmax} = 6000\,N$$

full thrust $\rightarrow \max(\delta_T) = 1$

$\gamma(V) = 0.252\,rad = 14.43°;$

d) The aircraft now flies a turn (see the data below). Which kinematic bank angle and what velocity is required for this maneuver?

Height and density	h / ρ	1000 m / 1.1117 $\frac{kg}{m^3}$
Turn rate	$\dot{\chi}$	$5\frac{°}{s}$
Lift coefficient during turn	$C_{L,turn}$	1

Definition of turn:

$\dot{V} = 0$

$\dot{\gamma} = 0$, $\gamma = 0$ } level flt

$\dot{\chi} = \text{const.}$

$T = D$

$L\sin\mu = mV\dot{\chi}$

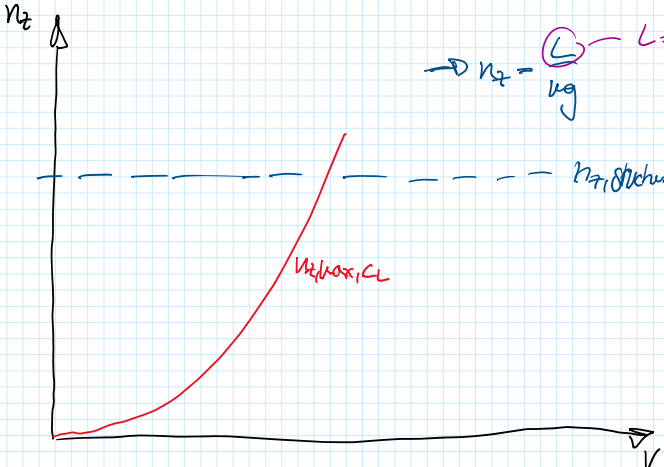
$L\cos\mu = mg$

important parameters: $\dot{\chi}$, n_z

$$(n_z)_{trim} = \frac{L}{mg} = \frac{1}{\cos\mu} \rightarrow (n_{z,horizontal})_{max} = 1$$

$$c_L = \frac{V}{\dot{\chi}}$$





$$\rightarrow n_z = \frac{L}{mg} \qquad L = mg\sqrt{1 + \left(\frac{V\dot{\chi}}{g}\right)^2} \Rightarrow n_z = \sqrt{1 + \left(\frac{V^2}{gr_c}\right)^2}$$

$$\dot{\chi} = \frac{g}{V}\sqrt{n_z^2 - 1}$$

$$r_c = \frac{V^2}{g\sqrt{n_z^2 - 1}}$$

(1) $L\cos\mu = mg$

(2) $L\sin\mu = mV\dot{\chi}$ $\rightarrow \tan(\mu) = \frac{V\cdot\dot{\chi}}{g}$ (3)

$L = \frac{1}{2}\rho V^2 S_{ref}\cdot c_L$ (4)

V

(1) $L\cos\mu = mg$

(2) $L\sin\mu = mV\dot{\chi}$ $\rightarrow \tan(\mu) = \frac{V\cdot\dot{\chi}}{g}$ (3)

$L = \frac{1}{2}\rho V^2 S_{ref}\cdot c_L$ (4)

(3) → (4): $L = \frac{mg}{\cos\mu} = \frac{1}{2}\rho V^2 S c_L$

⇒ 2 Unknowns: μ, V

$V = \frac{g\cdot\tan(\mu)}{\dot{\chi}}$

(3)+(5): $\frac{mg}{\cos\mu} = \frac{1}{2}\rho \frac{g^2\tan^2\mu}{\dot{\chi}^2} S\cdot c_L$

$mg = \frac{\rho\cdot g^2\cdot S c_L}{2\dot{\chi}^2}\frac{\sin^2\mu}{\cos\mu}$

$\sin^2\mu + \cos^2\mu = 1$

$\underbrace{mg}_{t} = \underbrace{\frac{\rho g^2 S c_L}{2\dot{\chi}^2}}_{s}\cdot\frac{(1-\cos^2\mu)}{\cos\mu}$

$t = s\frac{(1-\cos^2\mu)}{\cos\mu}$

⇒ $s\cdot\cos^2\mu + t\cdot\cos\mu - s \stackrel{!}{=} 0$

$x_{1,2} = \frac{-t \pm \sqrt{t^2+4s^2}}{2s}$

$x_1 = -1.8002$

$x_2 = 0.9257$ only one that makes sense

⇒ $\mu = \arccos(0.9257) = 0.3887\,rad = 22.27°$

$V = \frac{g\cdot\tan(\mu)}{\dot{\chi}} = 46\,m/s$